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Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

The table below shows the total costs faced by Allura's Little Robotics Company for different quantities of Good S sold.

Quantity	*Total Cost*
0	$100
1	$136
2	$180
3	$228
4	$280
5	$336
6	$396
7	$460
8	$528
9	$600
10	$680

Allura's Little Robotics Company sells Good S in a perfectly competitive market with a downward-sloping demand curve and an upward-sloping supply curve. The market price is $62 per unit.

(a) Calculate the average fixed cost of producing 2 units. Show your work.

(b) Identify the profit-maximizing quantity. Explain using marginal analysis.

(c) Calculate the economic profit at the profit-maximizing quantity you identified in part (b). Show your work.

(d) Based on your answer to part (c), will the number of firms in the industry increase, decrease, or stay the same in the long run? Explain.

(e) Based on your answer to part (c), will the market price increase, decrease, or stay the same in the long run? Explain.

(f) The income elasticity of demand for Good S is -0.5, and the cross-price elasticity of demand for toy robots with respect to the price of Good S is 0.8. Based on your answer to part (e), what will happen to the demand for toy robots? Explain.

(g) Now assume that the market in which Allura's Little Robotics Company operates is in long-run equilibrium at a price of $65 per unit.
(i) Suppose shipping costs per order for Allura's Little Robotics Company decrease. Will the profit-maximizing quantity

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of Good S for Allura's Little Robotics Company increase, decrease, or stay the same in the short run? Explain.
(ii) Instead suppose the government imposes a price ceiling of $\$60$ on the market for Good S. Will total economic surplus in the market for Good S increase, decrease, or stay the same in the short run as a result of the price ceiling? Explain.

1. Respond to all parts of the question.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response calculates the average fixed cost as $50 and shows the work.

$AFC = \frac{TFC}{Q} = \frac{\$100}{2} = \$50$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response identifies the profit-maximizing quantity as 6 units and explains that the 6th unit is the last unit where the marginal revenue ($62) is greater than the marginal cost ($60); the firm will not produce the 7th unit because the marginal cost of producing it ($64) is greater than the marginal revenue earned ($62).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response calculates the economic profit as $-\$24$ and shows the work.

$Economic\ Profit = TR - TC = 6(\$62) - \$396 = -\$24$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1 ✓

The response states that the number of firms in the industry will decrease and explains that negative economic profit will cause firms to exit the industry.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response states that the market price will increase and explains that with the exit of firms, the market supply will decrease, causing the price to increase.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response states that that demand for toy robots will increase and explains that the positive cross-price elasticity of demand for toy robots with respect to the price of Good S means that the goods are substitutes; therefore, an increase in the price of Good S will lead to an increase in the demand for toy robots.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response does both of the following:

· States that the profit-maximizing quantity of Good S will increase and explains that shipping costs per order are a variable cost, so a decrease in shipping costs per order will shift the marginal cost curve downward at each quantity so that marginal cost will intersect marginal revenue at a higher quantity.

· States that total economic surplus will decrease and explains that a price ceiling below the equilibrium price is binding

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and the quantity supplied will decrease, therefore decreasing total surplus.

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Anderson Company is a typical firm that manufactures Good G in a constant-cost, perfectly competitive market. Anderson Company is currently earning positive economic profit.

(a) What must be true about the relationship between accounting profit and economic profit if Anderson Company currently incurs both explicit and implicit costs in production?

(b) Draw correctly labeled side-by-side graphs for the market and Anderson Company and show each of the following.

(i) The market equilibrium price and quantity, labeled P_M and Q_M, respectively

(ii) The profit-maximizing price and quantity for Anderson Company, labeled P_F and Q_F, respectively

(iii) The area representing Anderson Company's positive economic profit, shaded completely

(c) On your graphs in part (b), show what will happen to each of the following if the market for Good G adjusts to long-run equilibrium.

(i) The market equilibrium price and quantity, labeled P_2 and Q_2, respectively

(ii) Anderson Company's profit-maximizing price and quantity, labeled P^* and Q^*, respectively

(d) Assume the production of Good G creates benefits for third parties.

(i) Given this situation, will the market equilibrium quantity be greater than, less than, or equal to the allocatively efficient quantity? Explain.

(ii) The government takes an action that corrects the externality in the market for Good G. As a result of the government's action, does total economic surplus increase, decrease, or stay the same? Explain.

2. Respond to all parts of the question.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

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The response accurately includes the criterion below.

- [ ] States that Anderson Company's accounting profit must be greater than its economic profit.

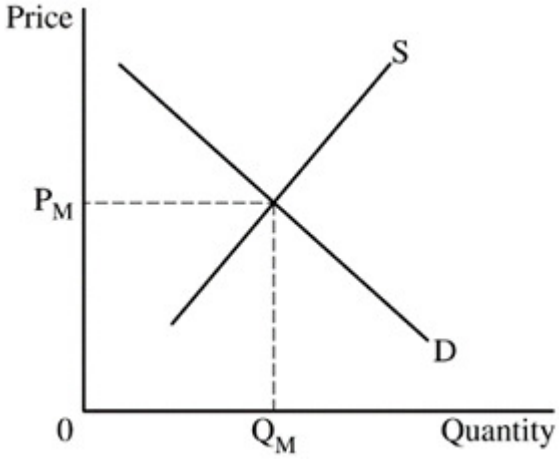
Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

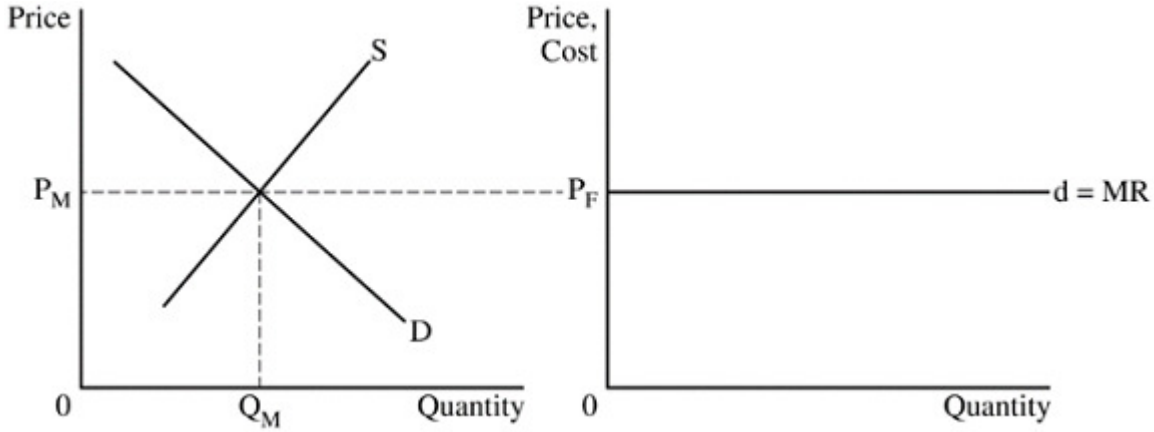
0	1	2	3	4	5 ✓

The response accurately includes five of the criteria below.

- [ ] Draws a correctly labeled graph of the market for Good G with a downward-sloping demand (D) curve and upward-sloping supply (S) curve and labels the market equilibrium price as $\mathrm{P_M}$ and the market equilibrium quantity as $\mathrm{Q_M}$.

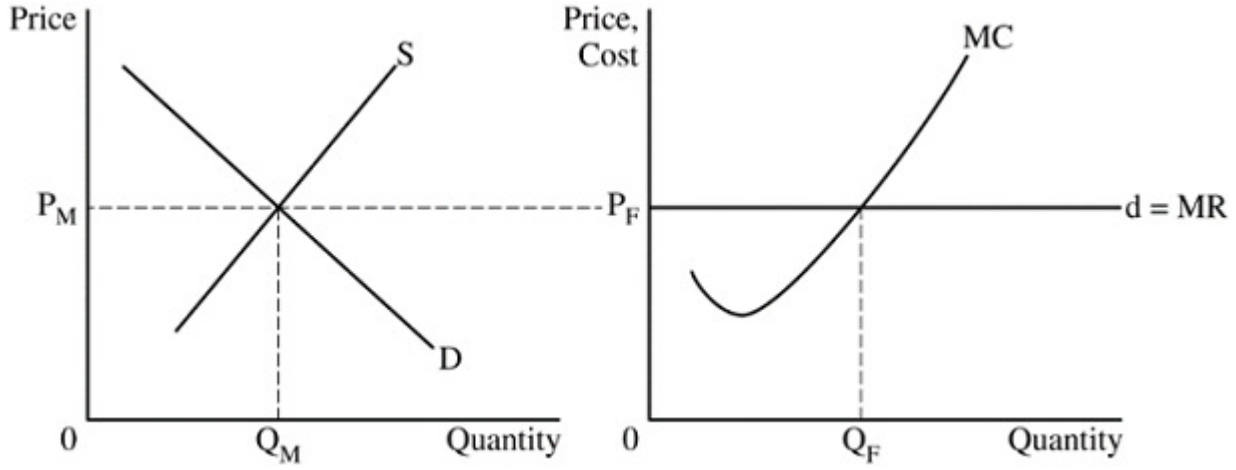


- [ ] Shows the firm's horizontal demand and marginal revenue ($\mathrm{d = MR}$) curve extended from the market equilibrium price ($\mathrm{P_M}$), and labels the firm's price as $\mathrm{P_F}$.

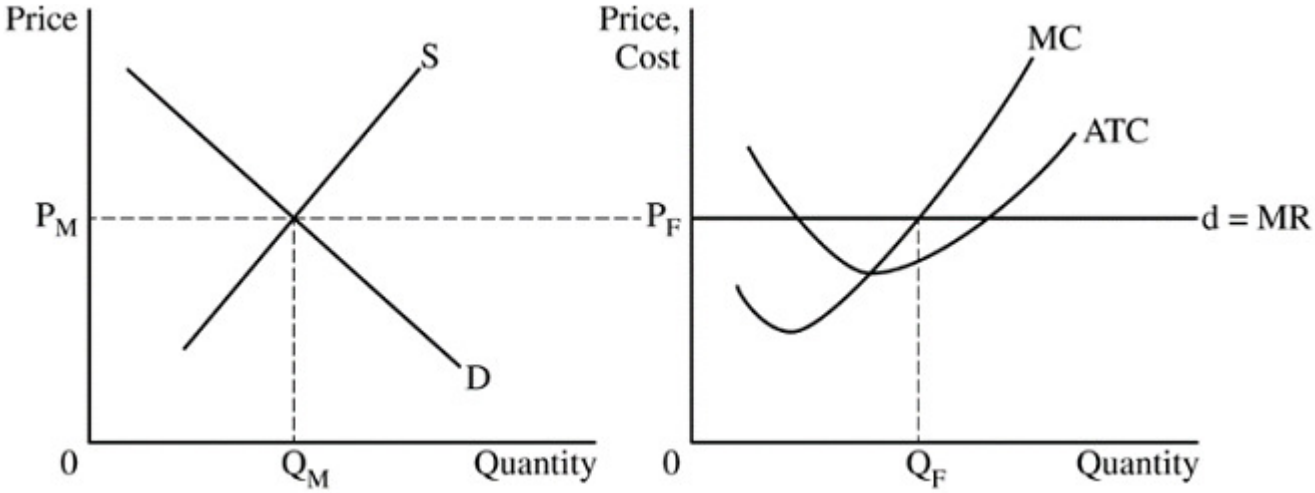


- [ ] Shows a rising marginal cost (MC) curve and the profit-maximizing quantity, labeled $\mathrm{Q_F}$ where $\mathrm{MR = MC}$.

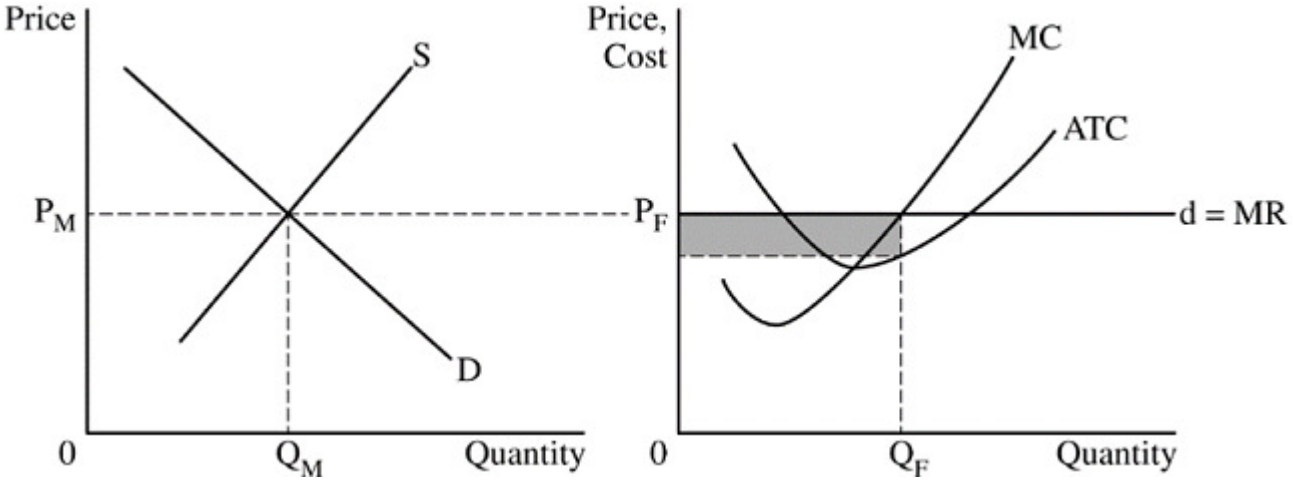
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- [ ] Shows the average total cost (ATC) curve below the firm's demand curve at Q_F and shows the MC curve passing through the minimum point of the ATC curve.



- [ ] Shows the area representing positive economic profit, shaded completely.



Part C

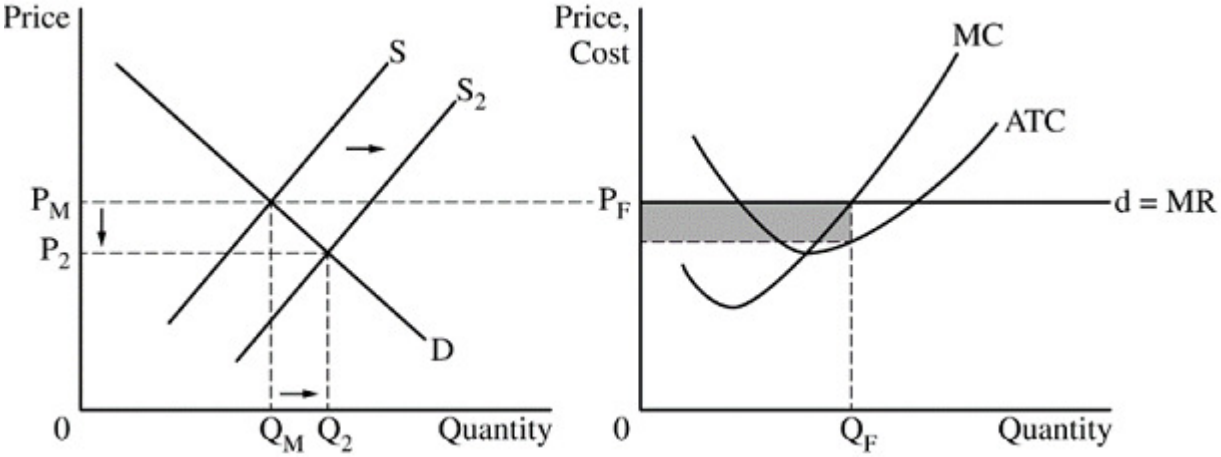
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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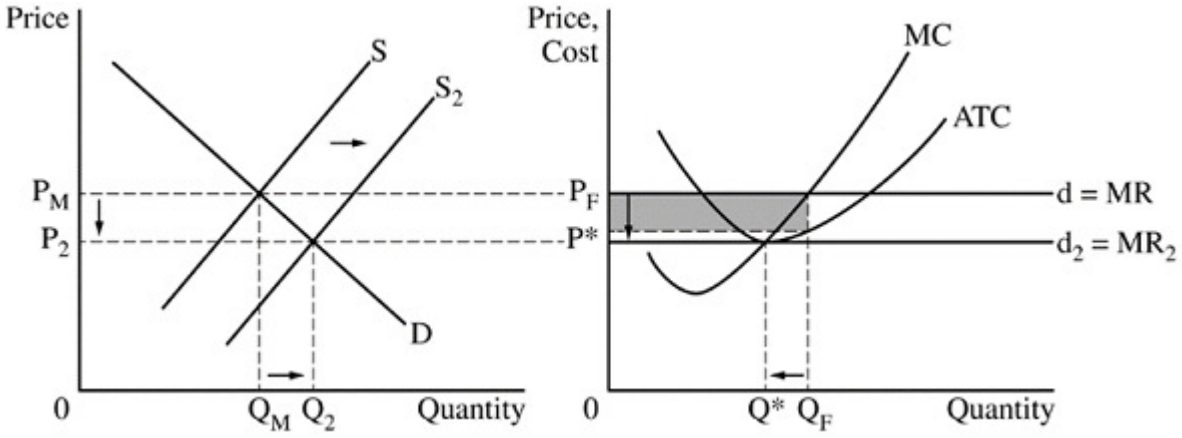
0	1	2 ✓

The response accurately includes two of the criteria below.

- ☐ On the market graph from part (a), shows a rightward shift in the market supply curve, resulting in a lower market equilibrium price, labeled P_2, and a greater market equilibrium quantity, labeled Q_2.



- ☐ Shows a lower price, labeled P^*, extended from the new market equilibrium price, P_2, and shows a lower quantity produced, Q^*, at the new intersection point of $P^* = MR_2 = MC = \text{minimum ATC}$.



Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

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The response accurately includes two of the criteria below.

- ☐ States that the market equilibrium quantity will be less than the allocatively efficient quantity and explains that the positive externality in production causes the marginal social cost to be less than the marginal private cost ($\text{MSC} < \text{MPC}$) at the market equilibrium.
- ☐ States that total economic surplus will increase and explains with ONE of the following.
  - The quantity produced will increase to the allocatively efficient quantity.
  - Deadweight loss will decrease to $0.
  - The marginal private cost will equal the marginal social cost, causing the externality to be internalized.

Assume that in a perfectly competitive market, a firm's costs and revenues are

marginal cost = average variable cost at $20

marginal cost = average total cost at $30

marginal cost = average revenue at $25

a. How will this firm determine the profit maximizing level of output?

b. What price will this firm charge? Explain how the firm determined this price.

c. Should this firm produce in the short run? Why or why not?

d. Will this firm earn a profit or incur a loss? Why?

3. Respond to all parts of the question.

Part A

1 Point

- **1 -** for MC=MR=P or MR=MC or P=MC

0	1 ✓

The student earns all of the following points:

1 - for MC=MR=P or MR=MC or P=MC

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Part B

2 Points

- **1 -** Price equals $25
- **1 -** Market/industry determines the price.

Or price may be used to link marginal revenue to average revenue.

0	1	2 ✓

The student earns all of the following points:

- **1 -** Price equals $25
- **1 -** Market/industry determines the price.

Or price may be used to link marginal revenue to average revenue.

Part C

1- for yes, and reason P>AVC or TR>TVC.

0	1 ✓

The student earns all of the following points:

1 - for yes, and reason P>AVC or TR>TVC.

Part D

1 - for loss and reason - P<ATC or TR<TC is acceptable.

0	1 ✓

The student earns all of the following points:

1 - for loss and reason - P<ATC or TR<TC is acceptable.

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Brody's firm produces trumpets in a perfectly competitive market. The table below shows Brody's total variable cost. He has a fixed cost of $240, and the price per trumpet is $60.

Quantity	Total Variable Cost
6	$120
7	$145
8	$165
9	$220
10	$290
11	$390

4. Respond to all parts of the question.

a. Calculate the average total cost of producing 6 trumpets. Show your work.

b. Calculate the marginal cost of producing the 11th trumpet.

c. What is Brody's profit-maximizing quantity? Use marginal analysis to explain your answer.

d. At the profit-maximizing quantity you determined in part (c), calculate Brody's profit or loss. Show your work.

e. Brody also produces saxophones at a loss in a perfectly competitive market. Draw a correctly labeled graph for Brody's firm showing the following at a market price of $200.

i. Brody's profit-maximizing quantity of saxophones

ii. Brody's loss, completely shaded

Part A

1 point is earned for: correctly calculating the ATC of producing 6 trumpets as $60 and for showing the work.

ATC = TC / Q = (FC + VC) / Q = ($240 + $120) / 6 = $360 / 6 = $60

OR

ATC = AFC + AVC = ($240 / 6) + ($120 / 6) = $40 + $20 = $60

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0	1 ✓

The student response earns all of the following points:

1 point is earned for: correctly calculating the ATC of producing 6 trumpets as $60 and for showing the work.

ATC = TC / Q = (FC + VC) / Q = ($240 + $120) / 6 = $360 / 6 = $60

OR

ATC = AFC + AVC = ($240 / 6) + ($120 / 6) = $40 + $20 = $60

Part B

1 point is earned for: correctly calculating the marginal cost of the 11th trumpet as $100.

0	1 ✓

The student response earns all of the following points:

1 point is earned for: correctly calculating the marginal cost of the 11th trumpet as $100.

Part C

1 point is earned for: correctly stating that Brody's profit-maximizing quantity is 9 trumpets, and for explaining how Brody uses marginal analysis. Brody determines quantity by equating MR to MC **OR** by choosing the greatest quantity with MR > MC: At the 9th trumpet: MR > MC ($60 > $55) and at the 10th trumpet: MR < MC ($60 < $70)

OR At the 9th trumpet: MR > MC ($60 > $55) and this is the last trumpet with MR > MC

Note: Using P instead of MR is also acceptable because P = MR for perfectly competitive firms.

0	1 ✓

The student response earns all of the following points:

1 point is earned for: correctly stating that Brody's profit-maximizing quantity is 9 trumpets, and for explaining how Brody uses marginal analysis. Brody determines quantity by equating MR to MC **OR** by choosing the greatest quantity with MR > MC: At the 9th trumpet: MR > MC ($60 > $55) and at the 10th trumpet: MR < MC ($60 < $70)

OR At the 9th trumpet:MR > MC ($60 > $55) and this is the last trumpet with MR > MC

3.7 perfect competition FRQ

Note: Using P instead of MR is also acceptable because P = MR for perfectly competitive firms.

Part D

1 point is earned for: correctly calculating Brody's profit as $80 and for showing the work.

Profit = Total Revenue – Total Cost = ($60 × 9) – ($220 + $240) = $540 – $460 = $80

OR

Profit = (P – ATC) × Q = ($60 – $51.11) × 9 = $80

0	1 ✓

The student response earns all of the following points:

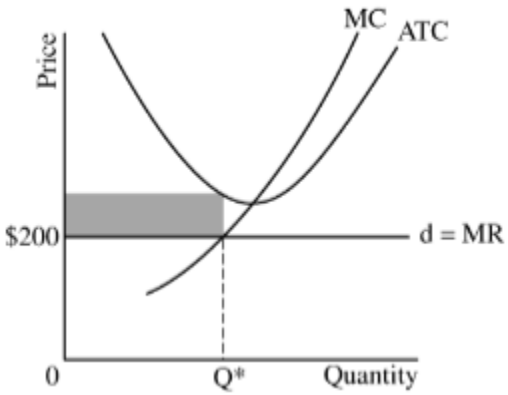
1 point is earned for: correctly calculating Brody's profit as $80 and for showing the work.

Profit = Total Revenue – Total Cost = ($60 × 9) – ($220 + $240) = $540 – $460 = $80

OR

Profit = (P – ATC) × Q = ($60 – $51.11) × 9 = $80

Part E



1 point is earned for: drawing a correctly labeled graph for a perfectly competitive firm with a horizontal demand curve at P = $200.

1 point is earned for: showing the profit-maximizing quantity, Q*, on the horizontal axis at MR = MC, connected with a dashed line.

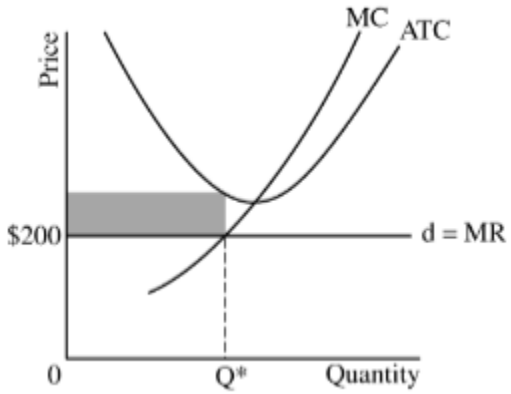
1 point is earned for: showing the ATC curve above the demand (d) curve at Q* and with the MC curve cutting the ATC curve at its minimum and for completely shading the area representing the loss.

Note: All quantities and prices should be labeled on the axes and connected to the intersection points by dashed lines.

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0	1	2	3 ✓

The student response earns all of the following points:



1 point is earned for: drawing a correctly labeled graph for a perfectly competitive firm with a horizontal demand curve at P = $200.

1 point is earned for: showing the profit-maximizing quantity, Q*, on the horizontal axis at MR = MC, connected with a dashed line.

1 point is earned for: showing the ATC curve above the demand (d) curve at Q* and with the MC curve cutting the ATC curve at its minimum and for completely shading the area representing the loss.

Note: All quantities and prices should be labeled on the axes and connected to the intersection points by dashed lines.